Quantification: I534_z19c1-2
============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	20
interior lacuna count	20
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 20)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	2815	2632	1594.62	px^2
major_axis_length_px	lacuna shape	86.4139	84.848	27.3025	px
minor_axis_length_px	lacuna shape	46.6552	44.321	13.8424	px
aspect_ratio	lacuna shape	1.9226	2.0925	0.5597	unitless
eccentricity	lacuna shape	0.7828	0.8784	0.1736	unitless
solidity	lacuna shape	0.745	0.7525	0.0994	unitless
roots_count	headline	10.9	10.5	4.5757	count
ring_length_r30_px	headline	281.35	272	101.823	px
ring_length_r60_px	reported	559.55	607.5	206.393	px
owned_length_px	ownership-dependent	685.131	469.268	604.064	px
edge_count	ownership-dependent	32.5	20.5	28.3131	count
mean_edge_length_px	ownership-dependent	21.1731	20.4438	3.4119	px
perimeter_px	normalised	288.247	279.099	98.758	px
ring_area_r30_px2	normalised	9678.3	9327	2520.39	px^2
ring_area_r60_px2	normalised	20407.5	21366	4636.47	px^2
in_frame_fraction_r30	normalised	0.9977	1	0.0105	unitless
in_frame_fraction_r60	normalised	0.9865	1	0.0417	unitless
ring_density_r30	normalised	0.0294149	0.0277583	0.0103823	px^-1
ring_density_r60	normalised	0.0273223	0.0255027	0.00887657	px^-1
roots_per_100px_perimeter	normalised	3.9421	3.2555	1.6358	per 100 px
ring_attached_length_r30_px	network v2	253.5	248.5	93.589	px
ring_attached_length_r60_px	network v2	401.25	417.5	185.494	px
ring_length_w_r30_px	network v2	319.521	313.028	116.23	px
ring_length_w_r60_px	network v2	635.254	691.319	234.232	px
sholl_crossings_r10	network v2	6.35	6.5	2.3232	count
sholl_crossings_r20	network v2	7.3	7.5	3.729	count
sholl_crossings_r30	network v2	7.05	7	3.1702	count
loop_count	loops, report only	1.45	0	2.9285	count
loop_edge_share	loops, report only	0.0823	0	0.118	share
ring_width_mean_r30_px	width	7.5417	7.5386	0.5354	px

Per field
----

measure	value	units
analysed_area_px2	845861	px^2
total_skeleton_length_px	21309	px
canalicular_length_density_per_px	0.0251921	px^-1
junction_count	412	count
junction_density_per_px2	0.00048708	px^-2
lacuna_count	20	count

lacunae_per_px2	2.364e-05	px^-2
skeleton_component_count	308	count
mean_component_length_px	69.1851	px
median_component_length_px	27	px
field_length_density_w_per_px	0.0285685	px^-1
field_density_without_flagged_per_px	0.0213652	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	418	px
unattached_component_count	24	count
canalicular_length_density_attached_per_px	0.0251921	px^-1
field_length_density_with_unattached_per_px	0.0256863	px^-1
excluded_area_px2	148874	px^2
analysed_field_area_px2	899702	px^2
loop_count	39	count
loop_edge_share	0.1195	share
edge_count	1205	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.033	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	21790	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1823	56.4889	49.8735	1.1326	0.4696	0.8271	-1.1579	125.003	757.529	False
2	1637	57.8693	48.4179	1.1952	0.5477	0.7276	-0.0784	295.51	896.101	False
3	1291	47.4215	38.1837	1.2419	0.593	0.8741	0.4227	324.111	336.189	False
4	6026	129.945	61.3358	2.1186	0.8816	0.8485	-1.1185	377.349	511.866	False
5	1342	66.6735	36.4005	1.8317	0.8378	0.6286	1.399	396.13	787.283	False
6	2978	75.2858	55.7926	1.3494	0.6714	0.7831	-0.8753	453.94	606.7	False
7	4876	84.2608	78.5547	1.0726	0.3617	0.7525	0.8631	503.31	62.7231	False
8	3737	84.848	59.7745	1.4195	0.7097	0.8338	-0.4377	493.364	441.851	False
9	820	51.4532	24.5895	2.0925	0.8784	0.7614	1.0688	505.329	211.1	False
10	3553	104.824	44.321	2.3651	0.9062	0.8721	0.2384	581.822	856.943	False
11	4782	122.937	62.4014	1.9701	0.8616	0.6511	1.4416	592.132	630.596	False
12	1615	90.9802	31.7451	2.866	0.9372	0.5886	-0.9716	606.27	724.946	False
13	2632	101.535	38.4077	2.6436	0.9257	0.6745	-1.3449	587.391	245.703	False
14	5166	140.866	61.0892	2.3059	0.9011	0.6428	0.013	686.162	69.5337	False
15	1712	87.747	36.8261	2.3827	0.9077	0.6786	-0.5403	724.265	671.19	False
16	914	54.9878	33.7201	1.6307	0.7899	0.5745	-0.9811	788.83	430.015	False
17	2692	86.6013	40.7838	2.1234	0.8822	0.8628	-1.1084	820.32	99.4658	False
18	1680	84.8432	32.3344	2.6239	0.9245	0.7391	-0.8417	827.977	860.805	False
19	4209	112.297	51.8973	2.1638	0.8868	0.8336	-0.6033	855.455	275.629	False
20								402.815	659.098	False

Network per lacuna

----

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	143	525	341.593	17	20.0937	7.2097
2	7	137	288	403.09	20	20.1545	6.609
3	9	269	718	2573.1	121	21.2653	6.7737
4	20	382	643	556.931	36	15.4703	7.0925
5	12	444	971	1741.59	84	20.7332	7.6615
6	9	177	200	241.965	13	18.6127	7.7118
7	16	421	656	883.465	37	23.8774	7.4938
8	15	237	506	307.764	17	18.1038	7.4688
9	13	385	605	755.097	48	15.7312	8.3586
10	15	328	887	1298.63	60	21.6439	8.1696
11	14	312	526	374.635	19	19.7176	7.9321
12	7	176	268	120.125	7	17.1607	7.936
13	17	368	607	475.86	21	22.66	7.4204
14	14	404	623	335.735	17	19.7491	7.9182
15	8	322	608	915.293	35	26.1512	7.1883
16	12	249	609	952.377	38	25.0625	7.5833
17	8	236	400	298.392	15	19.8928	6.9221
18	6	271	659	462.676	18	25.7042	8.3868
19	9	273	695	582.014	24	24.2506	6.8395
20	3	93	197	82.2843	3	27.4281	8.1584

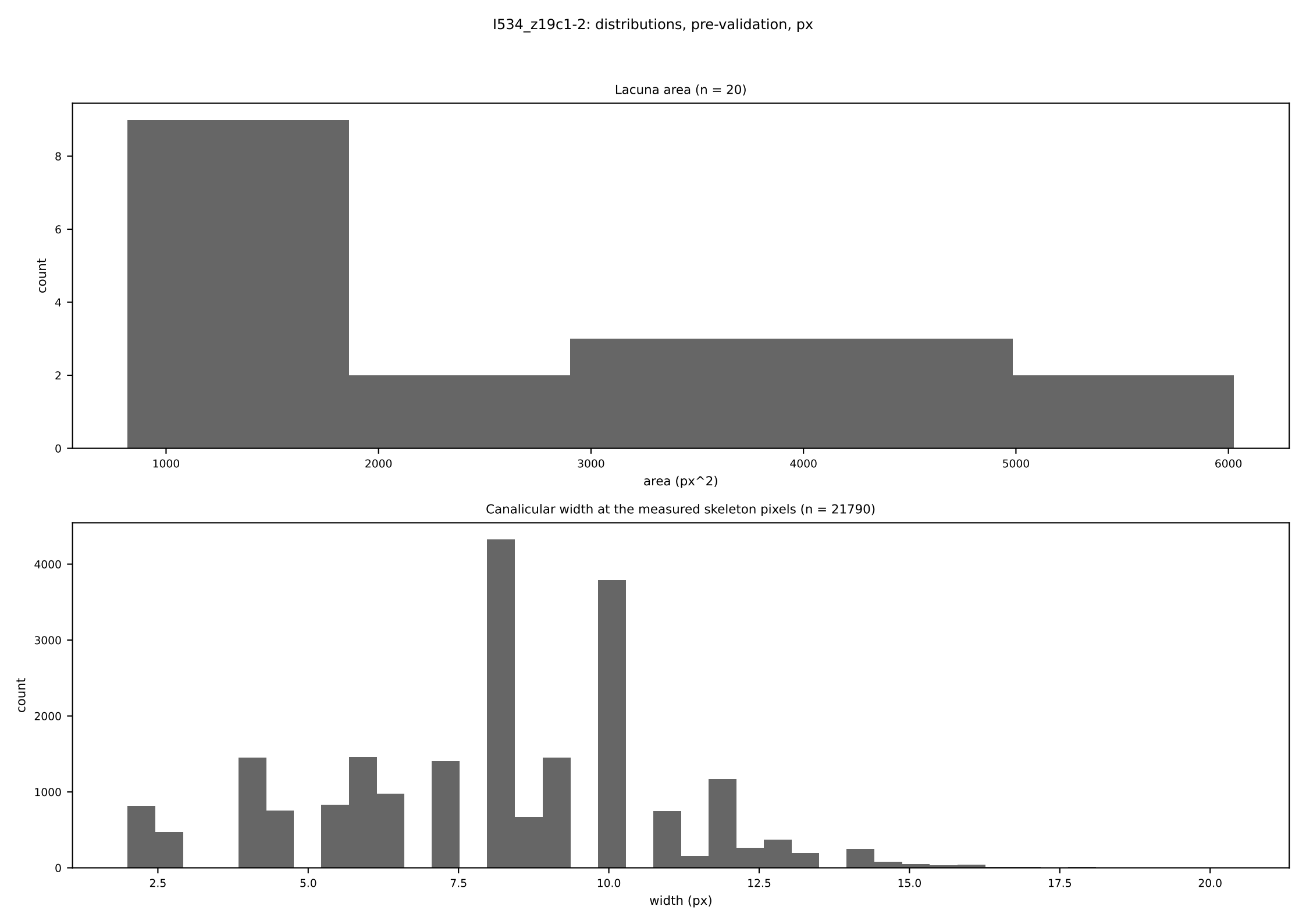
Normalised per lacuna
----

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	272.936	8396	22205	1	1	0.0170319	0.0236433	1.4655
2	283.463	8668	21456	1	1	0.0158053	0.0134228	2.4695
3	164.51	7220	19561	1	1	0.0372576	0.0367057	5.4708
4	382.854	12834	23401	1	1	0.0297647	0.0274775	5.2239
5	262.871	8654	20100	1	1	0.0513057	0.0483085	4.565
6	274.735	8747	13031	1	1	0.0202355	0.015348	3.2759
7	375.103	12127	21219	0.9532	0.8444	0.0347159	0.0309157	4.2655
8	300.049	10385	21896	1	1	0.0228214	0.0231092	4.9992
9	155.717	6688	13329	1	1	0.0575658	0.0453897	8.3485
10	288.735	10625	24527	1	1	0.0308706	0.0361642	5.1951
11	439.83	12113	21276	1	1	0.0257575	0.0247227	3.183
12	289.848	8213	14035	1	1	0.0214294	0.0190951	2.4151
13	316.777	10912	23620	1	1	0.0337243	0.0256986	5.3666
14	533.263	15650	25399	1	0.8882	0.0258147	0.0245285	2.6253
15	263.664	9930	21920	1	1	0.032427	0.0277372	3.0342
16	215.723	8071	20618	1	1	0.0308512	0.0295373	5.5627
17	247.279	9530	20421	1	0.9977	0.0247639	0.0195877	3.2352
18	251.22	9124	23689	1	1	0.0297019	0.0278188	2.3883
19	352.617	11633	27463	1	1	0.0234677	0.0253068	2.5523
20	93.7401	4046	8984	1	1	0.0229857	0.0219279	3.2003

Network v2 per lacuna
----

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	126	205	163.853	608.299	4	7	7
2	108	149	152.024	317.421	4	1	4
3	232	469	305.35	816.796	4	9	9
4	326	486	428.433	724.642	8	12	10
5	414	810	502.818	1100.51	10	13	12
6	169	180	202.167	226.894	3	4	1

7	409	615	486.931	745.612	10	11	10
8	222	282	265.794	568.973	6	5	4
9	341	541	434.635	679.487	8	12	7
10	304	679	365.007	996.666	7	9	13
11	286	345	346.664	597.6	6	3	5
12	176	256	199.995	306.735	4	3	4
13	317	405	418.647	691.512	8	10	7
14	343	430	460.546	711.583	9	11	10
15	287	439	376.676	700.269	8	9	8
16	249	486	288.836	691.127	5	4	5
17	199	250	270.007	457.517	6	8	7
18	248	518	298.309	729.6	8	7	8
19	238	390	320.706	811.193	7	7	8
20	76	90	103.012	222.652	2	1	2


I534_z19c1-2: distributions, pre-validation, px
Lacuna area (n = 20)
count
area (px^2)
0
2
4
6
8
1000
2000
3000
4000
5000
6000
Canalicular width at the measured skeleton pixels (n = 21790)
count
width (px)
0
1000
2000
3000
4000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0